

Krishi Attri

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Research Profile

Robotics researcher admitted to the Ph.D. programme at the University of Central Florida (Rehabilitation Engineering & Assistive Device Lab), currently completing an M.S. in Mechanical Engineering at Seoul National University under a Government Science Fellowship (GSFS). My research focuses on **online 3D reconstruction and pose tracking for robotic manipulation**, specifically developing visuo-tactile SLAM systems built around explicit **3D Gaussian Splatting** representations that enable robots to reconstruct and track objects under heavy occlusion during in-hand manipulation.

I am currently developing and integrating these systems as part of a humanoid robot project at the lab, fusing multimodal perception—RGB-D vision, DIGIT tactile sensing, and hand proprioception—into a shared Gaussian object state that simultaneously supports reconstruction, pose tracking, and manipulation interfaces. In parallel, I develop HiCache++, a training-free DMD velocity-cache that accelerates diffusion- and flow-based image-to-3D models—the same class of generative priors that seed my reconstruction pipeline. My work spans GPU-accelerated algorithm development, multimodal sensor fusion, and full-stack hardware–software integration on platforms including the UR5e and Allegro Hand.

Education

Ph.D., Mechanical Engineering — Post Master’s Track	Aug 2026 (Incoming)
University of Central Florida ORCGS Doctoral Fellow Rehabilitation Engineering & Assistive Device Lab (REAL) Advisor: Professor Hwan Choi	Orlando, FL, USA
Master of Science in Mechanical Engineering	Sept 2024 – June 2026 (Expected)
Seoul National University GSFS Scholar Soft Robotics & Bionics Laboratory	Seoul, South Korea
Bachelor of Science in Mechanical Engineering	Aug 2020 – May 2024
Villanova University Minor in Mechatronics Control & Dynamics  Diploma	Villanova, PA, USA
Exchange Student — B.S. Mechanical Engineering	Aug 2022 – June 2023
Yonsei University	Seoul, South Korea

Research Experience

Graduate Research Student	Sept 2024 – Aug 2026
Soft Robotics & Bionics Laboratory, Seoul National University	Seoul, South Korea
Humanoid Robot Project — “Alchemist” Project (MOTIE, Republic of Korea; Phase 2)	
- Contributing to Phase 2 (2024–2026) of the “Alchemist” Project — Korea’s national high-risk, high-reward R&D program (MOTIE; Phase 1 completed 2024) — at the Soft Robotics & Bionics Lab. - Leading integration of visuo-tactile SLAM and dexterous in-hand manipulation modules into the Phase 2 full-scale humanoid prototype. - Coordinating cross-disciplinary teams to hit quarterly hardware–software integration milestones.	

GaussianFeels: Object-Centric Gaussian SLAM for Visuo-Tactile In-Hand Manipulation (M.S. Thesis)
Advisor: Professor Yong-Lae Park, Seoul National University

- Developed GaussianFeels: an online visuo-tactile reconstruction and pose-tracking system built around an explicit object-centric 3D Gaussian Splatting map for robotic in-hand manipulation under heavy occlusion.
- Designed the system around four integrated components: (1) multimodal sensor ingestion synchronising RGB-D frames, segmentation masks, tactile contact geometry (DIGIT sensor), and hand proprioception into a per-frame representation; (2) an object-centric Gaussian map maintained online with contact-aware population management and occlusion-aware multimodal supervision; (3) a pose tracker embedded inside the SLAM loop supporting GT, GT-init, and full SLAM modes; and (4) a parallel manipulation branch for frame-0 shape completion and progressive geometry replacement.
- Implemented contact-aware Gaussian population management: multimodal Gaussian spawning, direct contact-point insertion, tactile-region density boosting, and selective freezing to preserve contact-critical regions under partial visibility.
- Designed an occlusion-aware training objective that dynamically reweights RGB, depth, and tactile supervision based on hand-induced occlusion, maintaining reconstruction fidelity when visual evidence becomes unreliable.
- Implemented a frame-0 shape-completion bootstrap using Hunyuan3D-2-mini: aligns an orientation-variant 3D prior to the real frame-0 observation, then progressively replaces generated geometry with measured camera and tactile data as the episode unfolds.
- Evaluated on the FeelSight benchmark family (14-object sweep) across simulation, real-world robot rollouts, and occlusion-focused episodes, benchmarked against the NeuralFeels and V-HOP baselines.
- **Results:** median ADD-S of 0.84 mm (simulation) and 3.4 mm (real) at ≈ 28 and ≈ 23.5 FPS on a single RTX 5090; retains 94% of simulation reconstruction F-score@5 mm on real hardware (0.95 $\rightarrow$ 0.89) versus 80% for NeuralFeels; matches or exceeds the model-free NeuralFeels baseline at $\approx 7.6\times$ the frame rate with no supplied CAD model.

PoP-SLAM: Point Cloud Projection for SLAM [Paper](#)

Co-authors: Seongmin Jung (SNU AI), Josselin Marchand (École des Mines), Michele Lorenzo Paolicchi (Vrije Universiteit Amsterdam)

- Co-developed PoP-SLAM: a dense visual SLAM system that introduces a projection-first rendering strategy, eliminating computationally intensive nearest-neighbor searches inherent to neural point cloud-based SLAM.
- Designed GPU-vectorised point cloud projection: transforms up to $\sim 15\,000$ neural points per frame into image space using vectorised matrix multiplication, enabling real-time SLAM at ~ 4 FPS.
- Introduced direct occlusion detection via multi-keyframe depth masking: projects neural points to nearby keyframes and retains only points consistent with measured depth across all keyframes, avoiding volume rendering entirely.
- Achieved **best ATE RMSE of 0.75 cm** on TUM-RGBD dataset, outperforming Point-SLAM, NICE-SLAM, ESLAM, and SplatAM; competitive 0.38 cm average on Replica dataset.
- Evaluated on synthetic (Replica) and real-world (TUM-RGBD) benchmarks; demonstrated robustness across varied indoor scenes.

Undergraduate Research Intern (Winter Internship)

January 2024

Soft Robotics & Bionics Lab, Dept. of Mechanical Engineering, Seoul National University [Letter](#) Seoul, South Korea

- Participated in a team project developing a surface electromyography (sEMG) sensor with flexible design elements for enhanced wearability and adaptability.
- Contributed to research on improving sensor stretchability through PDMS (polydimethylsiloxane) and vapour-deposited silver nanoparticles.
- Assisted in applying deep learning models (CNN-GRU, CNN-RNN, ViT) to sEMG signal processing, contributing to efforts to reduce classification error and improve gesture-recognition accuracy.
- Gained proficiency in ML-based signal processing and collaborated with a team of engineers to meet project milestones.

Robotics & Mechatronics Researcher

Villanova University ☉ [EOD robot](#) 📷 [Drone](#)

Aug 2023 – May 2024

Villanova, PA, USA

- Collaborated with a Ph.D. candidate on a dissertation titled “Autonomous Localisation and Navigation in GNSS-Denied Environments,” advancing SLAM algorithms via LiDAR–camera fusion and visual odometry for a quad-wheel robot.
- Developed a full ROS-based navigation stack for the robot, implementing path planning and obstacle-avoidance algorithms.
- Applied CNN-based feature extraction and point cloud generation from fused LiDAR–camera data to support autonomous decision-making without GPS.
- Integrated and configured Arduino and Raspberry Pi microcontrollers for robust ROS communication and real-time control.
- Implemented computer vision techniques (edge detection, landmark identification) to build real-time 3D environment maps.
- Implemented a 2D histogram localisation filter and a 1D Kalman filter tracker using probabilistic motion models for improved tracking accuracy.

Professional Experience

Indoor Farm Robotics Intern

Area2Farms ☉ [SILO demo](#)

June 2023 – Aug 2023

Arlington, VA, USA

- Contributed to the “Silo” automation tool for the local food system supply chain.
- Gained hands-on experience in extruded aluminium construction, pneumatics, industrial robotics, Arduino/Raspberry Pi programming, and agricultural irrigation systems.
- Collaborated with a cross-functional team to prototype automation solutions for indoor vertical farming.

Product Design Intern

Ampere LLC (Remote) 📄 [Certificate](#)

June 2022 – Aug 2022

Virtual, USA

- Contributed to 3D product modelling for upcoming consumer-technology designs, including **Shower Power** and an upcoming iPhone charging product.
- Analysed structural integrity and physics constraints to ensure product performance and durability.

Distance Education Operator

Villanova University

Sept 2021 – Dec 2021 & Aug 2023 – May 2024

Villanova, PA, USA

- Provided technical support for audio/video systems used in online course production.
- Managed live-stream recording and archiving of Distance Education materials for course content reliability.

Collections & Stewardship Technician

Villanova University

Jan 2022 – May 2022

Villanova, PA, USA

- Produced digital content for the university Digital Library using scanners, digital cameras, and archiving software.
- Managed records in cataloguing and acquisitions software; supported preservation of rare and special collections materials.
- Labelled and physically prepared new and rare library materials, and coordinated binding/repair shipments of books, periodicals, and theses.

Research Projects

GaussianFeels: Object-Centric Gaussian SLAM for Visuo-Tactile In-Hand Manipulation (M.S. Thesis)

Dec 2024 – Present

Objective: Achieve robust in-hand object reconstruction and pose tracking under heavy occlusion by

fusing RGB-D vision, DIGIT tactile sensing, and hand proprioception into a shared object-centric 3D Gaussian Splatting representation.

- **Core Representation:** Object-centric Gaussian map updated online from multimodal observations; stored in object coordinates and synchronised to world frame only when rendering or supervision requires it.
- **Population Management:** Contact-aware Gaussian spawning, density boosting at tactile contact regions, and selective freezing to preserve contact-critical geometry under partial visibility.
- **Pose Tracking:** SLAM-loop tracker combining tactile-aware frame-0 seeding, image-space depth sampling, and a multi-residual Levenberg–Marquardt SE(3) optimiser over a frozen Gaussian-density SDF (photometric, depth-SDF, tactile-SDF, tactile-normal, and one-sided non-penetration residuals; analytic Jacobian read through gsplat), with an ICP prior between consecutive depth observations.
- **Manipulation Branch:** Parallel process using Hunyuan3D-2-mini to generate a frame-0 shape prior; progressively replaced by measured camera and tactile geometry across the episode.
- **Evaluation:** FeelSight benchmark family (14-object sweep) — simulation, real-world rollouts, and occlusion-focused variants; benchmarked against NeuralFeels and V-HOP. Median ADD-S 0.84 mm (sim) / 3.4 mm (real) at ≈ 28 / ≈ 23.5 FPS on one RTX 5090; 94% sim-to-real F-score@5 mm retention versus 80% for NeuralFeels; $\approx 7.6\times$ NeuralFeels throughput with no supplied CAD model.

PoP-SLAM: Point Cloud Projection for SLAM (Seongmin Jung, Krishi Attri, Sept 2024 – Dec 2024 Josselin Marchand, Michele Lorenzo Paolicchi) [Paper](#)

- **Innovation:** Projection-first rendering pipeline projecting $\sim 15\,000$ learned neural points/frame on GPU via vectorised matrix multiplication — no nearest-neighbor search required.
- **Occlusion Handling:** Multi-keyframe depth masking selects only points consistent with measured depth across nearby keyframes — no volume rendering required.
- **Tracking Accuracy:** ATE RMSE 0.75 cm on TUM-RGBD (best among all evaluated methods including Point-SLAM, NICE-SLAM, ESLAM, SplaTAM); 0.38 cm average on Replica.
- **Speed:** ~ 4 FPS on NVIDIA RTX 4070 ($3\times$ faster than Point-SLAM baseline); $< 3.3\%$ overhead from point pruning.
- **Datasets:** TUM-RGBD (fr1-desk, fr1-room, fr2-xyz, fr3-office) and Replica (synthetic indoor scenes).

HiCache++: Training-Free DMD Velocity-Cache for Diffusion & Flow-Based 3D Generation 2026 [PyPI](#)

- Designed **HiCache++**, a model-agnostic, training-free DMD / exponential velocity-cache that skips redundant denoising compute in diffusion and flow samplers while preserving output fidelity — on Hunyuan3D-2.1 it holds reconstruction F-score where the polynomial basis collapses (0.86 vs 0.74 at skip-interval 5), extending the lossless skip range of the prior HiCache (Hermite) and TaylorSeer lines.
- Built and benchmarked acceleration across six-plus image-to-3D and generative backbones — Hunyuan3D-2 mini, Hunyuan3D-2.1, TRELIS v1 and TRELIS.2-4B, Meta SAM 3D Objects, Fast-SAM3D, and DiT-XL/2 (ImageNet) — benchmarking HiCache++ against the HiCache and TaylorSeer baselines.
- Released as a family of open-source repositories (Python; MIT / BSD); directly accelerates the Hunyuan3D image-to-3D prior used by the GaussianFeels frame-0 reconstruction branch.

splatreg & mathlas: Open-Source Research Tools 2026

- **splatreg** [GitHub](#) [PyPI](#) — a pure-PyTorch library that registers and merges native 3D Gaussian-Splatting scans into one SE(3)/Sim(3) frame via a closed-form-Jacobian Gaussian-SDF residual; matches GeoTransformer on official 3DMatch (91.5% recall) while adding the Sim(3) scale DoF it lacks, and reaches ADD-S AUC 0.995 on YCB object pose at ~ 17 ms.
- **mathlas** [GitHub](#) [PyPI](#) — an “airtight-math” MCP toolkit (13 LLM-free, data-returning tools) giving AI agents a 1.6M-document theorem search (R@10 0.998), PSLQ/OEIS constant & sequence identification, and Lean-kernel formal verification at a 0% false-positive rate; its self-augmenting retrieval loop surpasses TheoremSearch (59% vs 45% Hit@20).

Personal & Course Projects

Computer Vision Object Detection Web Application

Summer 2024

- Built a full-stack web application for real-time image and video object detection using a React frontend and Flask backend.
- Implemented and benchmarked five detection models: Faster R-CNN, Mask R-CNN, RetinaNet, Keypoint R-CNN, and SSDlite using PyTorch and TensorFlow with COCO dataset evaluation. Key findings: Mask R-CNN achieved highest annotation confidence (>0.8 scores); SSDlite fastest throughput; Faster R-CNN balanced both.
- Evolved through 8 development versions: from single-model photo processing through modularisation to multi-model video comparison, with an in-development custom “Focus Auto-Population Scan” model.
- Stack: Python (62%), JavaScript (23%), CSS (13%), HTML (1%).

Capstone: Plant Lifting Device for 3D Imaging [📺 Demo](#) [🏆 Award](#) [📷 Photos](#)

Aug 2023 – May 2024

Sponsored by a major agrochemical company — **1st Place, Most Innovative Solution**

- Led a multidisciplinary team through the full design-to-prototype lifecycle: actuator selection, motion system (belts, pulleys, drive gears), wiring, waterproofing, and plant platform engineering.
- Addressed sponsor constraints (strict size limits), plant stability risks, and motor-damage risks by developing a stabilising platform and controlled-speed mechanism.
- Applied project management across all design phases; awarded first place for Most Innovative Solution at the university capstone showcase.

Arduino Projects — Mechatronics Coursework

Jan 2022 – May 2022

- Designed and built three Arduino-based projects integrating pressure, photo, and sound sensors with actuators, motors, and LCD displays.
- Notable project: a puzzle box that randomised solution steps via algorithm and deployed a glitter-spray penalty for incorrect sequences.

Beetle-Bot — Miniature Combat Robot [📷 Photo](#)

Aug 2021 – Dec 2021

- Collaborated in a four-member team to build a miniature combat robot from scratch; achieved 3rd place among sophomore Mechanical Engineering students.

Previous Personal Projects [📷 Photos](#)

Summer 2021

- **Wi-Fi Drone:** Designed and programmed a Wi-Fi-controlled drone with custom stability logic.
- **Swarm Drones:** Developed multi-drone coordination algorithms for communication and synchronisation.
- **Robotic Arm:** Built a servo-actuated robotic arm programmed for pick-and-place operations.
- **3D Printer:** Assembled a 3D printer from scratch including hardware build and slicing software configuration.

SOLIDWORKS Project — Functional Scooter Model

Jan 2021 – May 2021

- Completed SOLIDWORKS coursework; collaborated in a four-member team to design and 3D-print a functional scooter model.

AI Mini-Project — Basketball Outcome Prediction

Aug 2020 – Dec 2020

- Explored ML fundamentals in Python; collaborated in a three-member team to build a predictive model for basketball game outcomes.

ICE Competition — Assistive Device for the Visually Impaired

Aug 2020 – Dec 2020

Innovation, Creativity & Entrepreneurship Competition (IIE Idea Challenge) — **3rd Place, College of Engineering**

- Led ideation and business model development in a four-member team, designing a wearable assistive device to aid visually impaired individuals.

Technical Skills

SLAM & 3D Reconstruction

- 3D Gaussian Splatting, Visuo-Tactile SLAM (GaussianFeels), Neural Point Cloud SLAM (PoP-SLAM)
- Point Clouds, Structure from Motion (SfM), RGB-D Reconstruction, Object-Centric Gaussian Mapping
- Contact-aware Gaussian population management, occlusion-aware multimodal supervision, Levenberg–Marquardt SE(3) optimisation, gsplat
- SLAM benchmarking (TUM-RGBD, Replica datasets)
- Point-cloud & Gaussian-splat registration: SE(3)/Sim(3), Gaussian-SDF residual, ICP, FPFH+RANSAC, GeoTransformer (3DMatch / 3DLoMatch); splatreg

Deep Learning & AI

- PyTorch, TensorFlow, NVIDIA CUDA (custom kernel development)
- Multimodal Learning, Differentiable Rendering
- CNN, RNN, CNN-GRU, CNN-RNN, Vision Transformer (ViT)
- ML for signal processing and predictive analytics
- Reproducible ML workflows, experiment pipeline design, model orchestration
- Ollama (local LLM inference), Genesis (robotics simulation)

Generative 3D & Diffusion Acceleration

- Training-free sampler acceleration: **HiCache++** (DMD / exponential velocity caching); HiCache (Hermite feature caching), TaylorSeer
- Image-to-3D & 3D generative models: Hunyuan3D-2 / 2.1, TRELIS v1 / v2, Meta SAM 3D, Fast-SAM3D, DiT
- 3D Gaussian Splatting registration & merging (splatreg), gsplat
- Spectral forecasting: Dynamic Mode Decomposition (DMD / Prony / Matrix-Pencil)

AI Agents, Retrieval & Formal Methods

- MCP (Model Context Protocol) tool & server development for AI agents (mathlas)
- Dense & hybrid retrieval: transformer embeddings (Qwen3), BM25, reciprocal-rank fusion
- Formal & numeric verification: Lean 4 kernel, PSLQ, OEIS, high-precision (mpmath) identification
- Program synthesis / evolutionary search (FunSearch), sandboxed evaluation

Robotics & Middleware

- ROS, OpenCV, Open3D, COCO API
- Sensor Fusion, Real-time Inference, Path Planning, Autonomous Navigation
- Simulation & Sim-to-Real: NVIDIA Omniverse → UR5e + Allegro Hand
- Robot hardware/software integration (perception ↔ actuation APIs)

Hardware & Sensors

- DIGIT Visuo-Tactile Sensor, RGB/RGB-D Cameras, LiDAR (camera fusion)
- VectorNav IMU, Emlid RTK GPS, NVIDIA GPUs
- UR5e Robotic Arm, Allegro Hand, Raspberry Pi / Teensy / Arduino
- 3D Motion Capture (Vicon/OptiTrack)
- 3D Modelling & Printing, Machine Shop Tools

Programming Languages

- Python, C/C++, CUDA, MATLAB, Arduino, $\text{\LaTeX}$, HTML, Django, MySQL, VHDL, Maple

Software & Tools

- Simulation & CAD: NVIDIA Omniverse, SOLIDWORKS, Blender, KiCAD
- Dev Tools: Git/GitHub, Jupyter Notebooks, Anaconda, Linux (Ubuntu), PowerShell
- Web: React.js, Flask

- Office & Misc: Microsoft Office, Matplotlib, Sklearn

Languages

- English (Fluent), Hindi (Fluent), Korean (Beginner)

Certifications, Licenses & Awards

NVIDIA Computer Vision Nanodegree

Winter 2024

Udacity in partnership with NVIDIA

- Modules: Introduction to Computer Vision, Advanced CV & Deep Learning, Object Tracking & Localisation, Cloud Computing, Neural Network Training, C++ Programming, Applied Deep Learning Projects.

Innovative Robot Technologies and their Applications [Certificate](#)

Aug 2024

K-MOOC × Seoul National University Credential ID: c65ee02c32ee88873985a799fab29ef8

GRE General Test [Report](#)

Sept 2023

Credential ID: 82736046 Blockchain ID: 809252

- Verbal Reasoning: 153 (56th percentile)
- Quantitative Reasoning: 166 (80th percentile)
- Analytical Writing: 4.0 (56th percentile)

Capstone Project — 1st Place, Most Innovative Solution [Award](#)

May 2024

Villanova University Capstone Showcase (Industry-Sponsored)

ICE Competition — 3rd Place

Dec 2020

Innovation, Creativity & Entrepreneurship Competition, College of Engineering

Beetle-Bot Competition — 3rd Place

Dec 2021

Villanova University Mechatronics Course

Dean's List [Proof](#)

Fall 2020, Spring 2021

Villanova University

UPenn Robotics Specialisation

Summer 2021

Coursera in partnership with University of Pennsylvania

- Aerial Robotics, Computational Motion Planning, Mobility, Perception, Estimation and Learning, Capstone.

Relevant Coursework

Seoul National University

- Deep Learning
- Topics in Control and Automation
- Smart Materials and Design
- Precision Metrology and Vision Inspection
- Control Systems 1
- Engineering Research Ethics and Writing Skills
- TA: Mechanical Product Design

Villanova University

- Computer Programming for Mechanical Engineering
- Mechatronics & Microcontrollers and Applications
- Dynamic Systems I & Lab
- Capstone Design I & II
- ME Undergraduate Research I
- Manufacturing Engineering

- Mechanics of Materials / Solid Mechanics & Design I
- Electrical Circuit Fundamentals
- Material Science I
- Thermodynamics
- Engineering Interdisciplinary Project I
- Intro to Computer-Aided Design & Craft (SOLIDWORKS)
- Statics; M.E. Analysis & Design; ME Laboratory II
- Fundamentals of Computer Programming Engineering I & Lab

Yonsei University (Exchange)

- Heat Transfer, Fluid Mechanics, Mechanical Vibrations, Mechanical Systems Control, Basic Circuit Theory, Probability and Random Variables

Leadership & Activities

Secretary

Villanova CubeSat Club

May 2021 – May 2023

Villanova University

- Fostered community engagement in space science with a focus on CubeSats.

Representative & Head of Special Events

Villanova International Students' Organisation (VISO) [📄 Certificate](#)

Aug 2020 – Aug 2022

Villanova University

- Built connections between upperclassmen and incoming international first-year students; organised social events to strengthen community.

Member

Society of Asian Scientists and Engineers (SASE)

May 2021 – Aug 2022

Villanova University

- Supported SASE's mission of preparing Asian-heritage scientists and engineers for the global business landscape.

Member

American Society of Mechanical Engineers (ASME)

May 2021 – Aug 2022

Villanova University

- Contributed to professional development activities and chapter engineering initiatives.